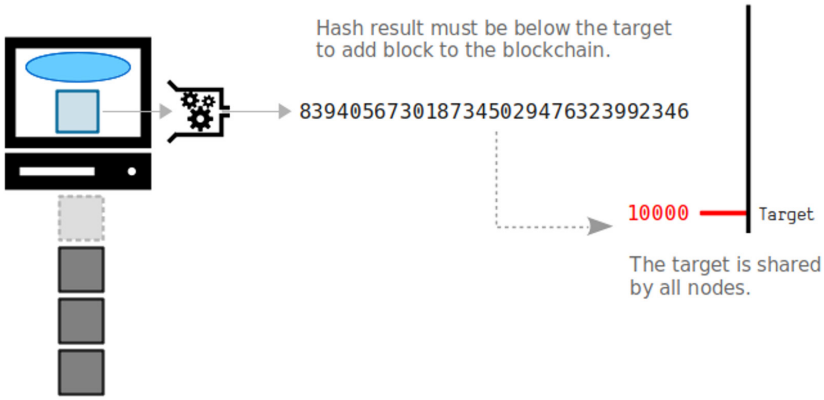
While a block hash **can be used to verify the integrity** of the data in the block, it **does not reveal all of the information** within the block. The information within the block can only be accessed by using the cryptographic keys required to decode it. The block hash simply provides a means of verifying that the data in the block has not been altered.

The Race to Mine a Block

Miners engage in a competition to uncover the block hash that aligns with the target (a special number) set by the network. The miner who successfully discovers the correct block hash is granted the opportunity to add that block to the blockchain and assign it with the corresponding hash ID. This solution serves as validation for the block's authenticity.



- Mining can be compared to a race where the goal is to reach the finish line as quickly as possible. The **difficulty target** in the race is adjusted periodically, making it harder to mine a block as more miners join the race.

- Let's say the target set by the network in a *blockchain* is 1000. The miners would have to use their computers to search for a special number, called the block hash, that is lower than 1000. The first miner to find a block hash that is lower than 1000 gets to add a group of transactions to the blockchain and is rewarded with some ***bitcoin***.

The **difficulty level** is a measure of how difficult it is to find a valid block hash that meets the target set by the network. It is adjusted periodically to ensure that blocks are added to the blockchain at a consistent rate. The difficulty level is expressed as a number, and the higher the difficulty level, the more difficult it is to find a block hash that meets the target.

- For example, consider two different hashes:
  - Hash 1: **0000A1mINgF0RbL0cK5wItHth3hAy5tAcK**
    Difficulty level: 1

  - Hash 2: **00000000A1mINgF0RbL0cK5wItHth3hAy5tAcK**
    Difficulty level: 2